TdRCB User Manual

TdRCB: An open-source software for estimating tempera ture-dependent receiver code bias variations

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1 Overview

Multi-system and Multi-frequency Receiver Code Bias Analysis Software (TdRCB) is an open-source GNSS data processing package specifically designed for extracting time-varying receiver code biases (RCBs). TdRCB is developed in C/C++, which can be easily ported to different operating systems, such as Windows and Linux. It is a post-processing software, which can process multi-frequency data from GPS, Galileo, BDS, GLONASS. Using a least-squares filter (LSF) based on undifferenced and uncombined observations to estimate time-varying RCBs. Furthermore, it requires observation files and broadcast ephemeris, offering a simple operation process. Processing strategies for different data can be configured in the formatted configuration file.

2 TdRCB software

2.1 Structure of TdRCB

Generally, the processing can be summarized into three parts.

（1）**Data preparation.** The required input files include the observation file and the broadcast ephemeris file. The processing parameters should then be specified in the configuration file. Temperature data are optional.

（2）**Data processing.** In this step, TDRCB first calculates the satellite information. Cycle slip detection is then performed using the geometry-free combination and the Melbourne-Wübbena combination. After finishing cycle slip detection, the LSF is employed to estimate unknown parameters. Finally, quality control is performed using the DIA method.

（3）**Information output.** In this step, processing information and estimated parameters, including satellite clock offsets, satellite elevation and azimuth angles, post-fit code and carrier-phase residuals, and time-varying receiver code biases (RCBs), are written to formatted output files. The log file primarily records satellite exclusions during data processing. If temperature data are provided, the software also outputs the fitted model describing the relationship between RCBs and temperature, along with the corresponding fitting residuals.

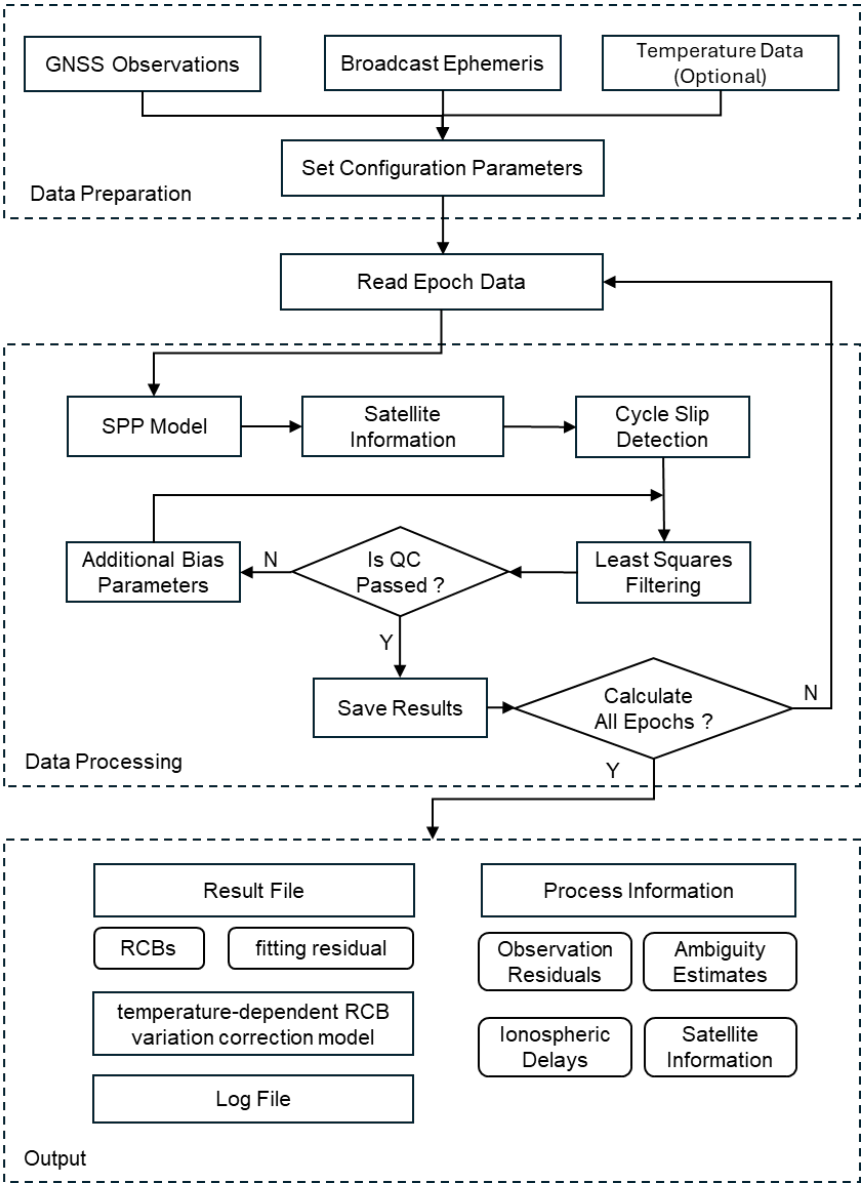


Fig. 1 The processing framework of TdRCB software

2.2 Processing strategies

For the processing strategies. The frequency-independent parameters, including the range, receiver clock offsets, satellite clock offsets, and tropospheric delays, were grouped into a single parameter and estimated as white noise. Ionospheric delays, ambiguities, and time-varying receiver code biases (RCBs) were also estimated. The detailed processing strategies are summarized in Table 1, and the complete mathematical derivations are provided in the Appendix for reference. Users can adjust the processing strategies according to their specific requirements.

Table 1 Processing strategies of TdRCB

Item	Processing strategies
Satellite constellation	GPS/GLONASS/BDS/Galileo/QZSS
Frequency number	Dual-frequency and above (up to 5 frequencies)
Functional mode	Uncombined model
Sampling rate	30s
Cut-off mask angle	10°
Observation weight	$P = \sin^2(E) / \sigma 2$($\sigma 2$ indicates variance of code or carrier phase) A priori precision of 0.003 and 0.3 m for raw phase and code
Filtering method	Least square filter
Frequency-independent parameters	Estimated as white noise
Ionospheric delays	Estimated as white noise
Phase ambiguities	Estimated as float constants for each satellite and frequency
SCB	Estimated as day constants for each satellite on the third or higher frequency band
RCB	Estimated as white noise
IFB	GLONASS, Estimated as white noise

Table 2 lists the frequency priority order implemented in the code.

Table 2 Frequency priorities for each system

GNSS Frequency	1	2	3	4	5
GPS	P1	P2	P5	\	\
GLONASS	R1	R2	\	\	\
Galileo	E1	E5a	E5b	E6	E5
QZSS	P1	P2	P5	\	\
BDS-2	B1I	B2I	B3I	\	\
BDS-3	B1I	B3I	B2b	B1C	B2a

3 Requirements

3.1 Supported platforms

TdRCB software was developed in the C/C++ language using the cross-platform CMAKE build system. It can be compiled and executed on both Windows and Linux operating systems. It is recommended to compile and debug TdRCB under Visual Studio 2022 in Windows or using GCC/Clang in Linux. Computer configuration requirements are as follows:

Operating system: Linux or Windows

System type: 32 or 64 bit

Memory: at least 512MB

Hard disk space: at least 500MB

CMAKE Version: CMAKE version higher than 3.10

Visual Studio Version: at least version 2022

The software has been tested under Windows 10 and Ubuntu 16.04 or higher, with all tests passed.

3.2 License

Copyright (C) 2026 by APM, CAS, All rights reserved.

This program is an open-source software: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation.

4 File structure

The project directory structure is as follows.

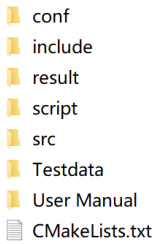


Fig. 2 File list

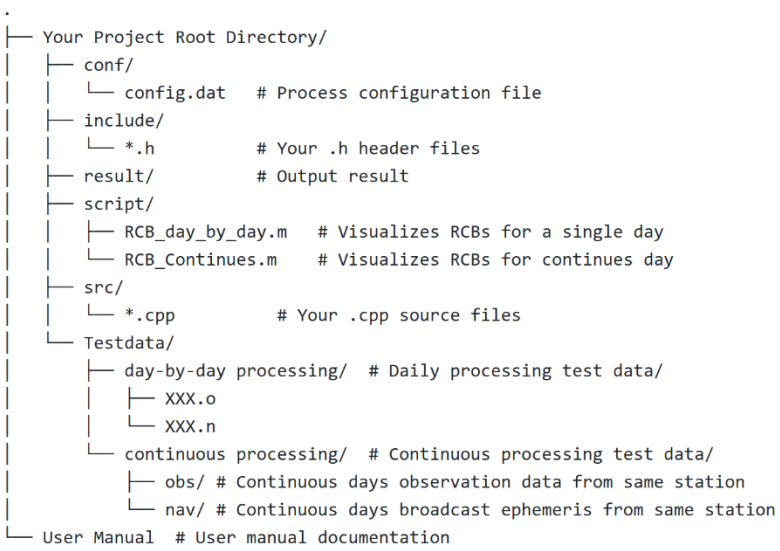


Fig. 3 The structure of File

5 Configuration File (config.dat)

The configuration file mainly contains four parts:

a) Input file configuration

b) Output file configuration

c) Processing strategies configuration

d) Station names

5.1 Input file configuration

Users should specify the input file paths in conf/config.dat. All paths are interpreted relative to the current working directory from which TDRCB is executed. Depending on the processing mode, users should configure either individual file paths or directory paths. The applicable parameters are controlled by Continue:

- Continue = 0 (daily processing): Use OBS_PATH and NAV_PATH.
- Continue = 1 (continuous/multi-day processing): Use OBS_DIR and NAV_DIR.

（1）OBS_PATH

Specifies the path to a single-station RINEX observation file. This parameter is used only when Continue=0.

（2）NAV_PATH

Specifies the path to the broadcast ephemeris file. This parameter is used only when Continue = 0.

（3）TEMP_PATH

Specifies the path to the temperature data file. This parameter is optional. If left empty, temperature-related

processing is skipped, and the .tfit and .tres output files are not generated. The temperature data file must meet the following requirements:

- File format: CSV text file, with columns separated by commas or tabs.
- Column order: Year, month, day, hour, minute, second, and temperature.
- Temperature unit: Degrees Celsius (°C).
- Sampling interval: 30 s, covering a complete 24-hour period (2,880 epochs).
- Missing values: Missing temperature values can be left blank and will be interpolated by the software.

（4）OBS_DIR

Specifies the directory containing multi-day observation files. This parameter is used only when Continue = 1. The software scans this directory for observation files.

（5）NAV_DIR

Specifies the directory containing broadcast ephemeris files for multiple days. This parameter is used only when Continue = 1. The software scans this directory for broadcast ephemeris files.

```
#====================================================================
# 1. Input
# One observation file and one navigation file are used.
#====================================================================
# OBS_PATH   RINEX observation file
# NAV_PATH   broadcast ephemeris
# TEMP_PATH  optional temperature file. Leave empty to skip.
OBS_PATH  =..\Testdata\day-by-day processing\LONA00AUS_R_20202690000_01D_30S_MO.20o
NAV_PATH  =..\Testdata\day-by-day processing\brdc2690.20n
TEMP_PATH =..\Testdata\day-by-day processing\lona_tem_2020269.csv

#====================================================================
# Multi-day
# To switch: set Continue = 1, comment out OBS_PATH and NAV_PATH,
#====================================================================
OBS_DIR   =..\Testdata\continuous processing\obs\
NAV_DIR   =..\Testdata\continuous processing\nav\
```

Fig. 4 Input file configuration

5.2 Output file configuration

Users should specify the output paths in conf/config.dat. All paths are interpreted relative to the current working directory from which TDRCB is executed.

（1）OUT_PRO

Specifies the directory for intermediate processing information. The output files use the .pro extension. The files contain information such as satellite positions, satellite clock offsets, ambiguities, elevation angles, and

azimuth angles. For details, see Section 7.1.

（2）OUT_PATH

Specifies the directory for the main output files. The following three types of files are written to this directory:

Extension	Description
.out	Epoch-wise receiver code bias (RCB) estimates.
.tfit	Quadratic fitting model describing the relationship between temperature and RCB. Generated only when TEMP_PATH is configured.
.tres	Fitting residuals, calculated as the estimated RCB minus the model-predicted RCB. Generated only when TEMP_PATH is configured.

（3）OUT_LOG

Specifies the directory for log files, which use the .log extension. The log files record satellites excluded during processing and the corresponding reasons for exclusion, such as insufficient satellite elevation or an insufficient number of available frequency observations.

```
#====================================================================
# 2. Output
#====================================================================
# OUT_PRO    satellite detail (.pro)
# OUT_PATH   RCB solution (.out); also temperature fit (.tfit) and fit residual (.tres)
# OUT_LOG    rejected-satellite log (.log)
# Directory must already exist (program does not create it).
OUT_PRO   =..\result\
OUT_PATH  =..\result\
OUT_LOG   =..\result\
```

Fig. 5 Output file configuration

5.3 Processing strategies configuration

（1）UseBDS

Value	Description
0	Use BDS-3 satellites only.
1	Use BDS-2 satellites only.
2	Use both BDS-2 and BDS-3 satellites. The first two frequencies are standardized as B1I and B3I for both systems.

（2）Frequency

Specifies the number of frequencies used in processing. For example, 2 indicates dual-frequency processing.

（3）Exclsats

Specifies the PRNs of satellites to be manually excluded from processing. Multiple satellite PRNs should be separated by spaces.

（4）System

Specifies the GNSS constellations included in the solution.

（5）Continue

Specifies the processing mode:

- 0: Daily processing.
- 1: Continuous or multi-day processing.

```
#=============================================================================
# 3. Processing
#=============================================================================
# Ephopt      0 broadcast ephemeris
# UseBDS      0 BDS-3, 1 BDS-2, 2 BDS-2+BDS-3 (first two freqs: B1I B3I)
# Frequency   number of frequencies
# Elmask      elevation mask (deg)
# Exclsats    excluded PRNs, separated by spaces
# System      bit mask: GPS 1, GLO 4, GAL 8, QZSS 16, BDS 32
# Continue    0 day-by-day, 1 multi-day
Ephopt      =0
UseBDS      =2
Frequency   =2
Elmask      =10
Exclsats    =C01 C02 C03 C04 C05
System      =45
Continue    =1
```

Fig. 6 Processing options 1 configuration

5.4 Station Information

```
#=============================================================================
# 4. Station
#=============================================================================
StaName     =MRO1
DOY         =2024298
```

Fig. 7 Station information configuration

6 Compile and Run

6.1 Compile

6.1.1 Windows

Ensure that the complete compilation tool and CMAKE (version 3.10 or higher) is installed.

➢ Extract the files downloaded from GitHub

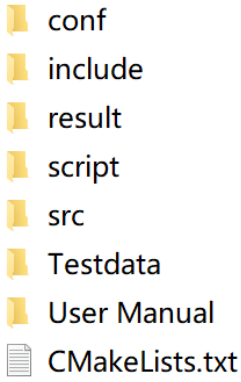


Fig. 8 The extracted files

- Search "x64 Native Tools Command Prompt for VS 2022"

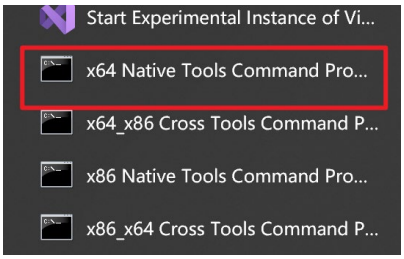


Fig. 9 Open Command Prompt

- Navigate to Project Directory and Compile. Execute these commands in sequence

① cd your_project_path ②mkdir build ③cmake -B build ④cmake --build build

```
C:\>cd C:\Users\Administrator\Desktop\test   (1)

C:\Users\Administrator\Desktop\test>mkdir build  (2)

C:\Users\Administrator\Desktop\test>cmake -B build (3)
-- Building for: Visual Studio 17 2022
-- Selecting Windows SDK version 10.0.26100.0 to target Windows 10.0.19045.
-- The CXX compiler identification is MSVC 19.41.34123.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: D:/VS2022/VS2022/VC/Tools/MSVC/14.41.34120/bin/Hostx64/x64/cl.exe - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (3.7s)
-- Generating done (0.0s)
-- Build files have been written to: C:/Users/Administrator/Desktop/test/build

C:\Users\Administrator\Desktop\test>cmake --build build  (4)
适用于 .NET Framework MSBuild 版本 17.11.9+a69bbaaf5

  1>Checking Build System
  Building Custom Rule C:/Users/Administrator/Desktop/test/CMakeLists.txt
  TdRCB_ComFun.cpp
  TdRCB_DataPre.cpp
  TdRCB_DataProcess.cpp
  TdRCB_Ephemeris.cpp
  TdRCB_Inpfile.cpp
  TdRCB_Options.cpp
  TdRCB_Pntpos.cpp
  TdRCB_RcbProcess.cpp
  TdRCB_ReadRinex.cpp
  TdRCB_SolBody.cpp
  TdRCB_SolHead.cpp
  main.cpp
  正在生成代码...
  TdRCB.vcxproj -> C:\Users\Administrator\Desktop\test\build\Debug\TdRCB.exe
  'pwsh.exe' 不是内部或外部命令，也不是可运行的程序
  或批处理文件。
  Building Custom Rule C:/Users/Administrator/Desktop/test/CMakeLists.txt
```

Fig. 10 Finish compile under Windows

6.1.2 Linux

Ensure that the complete compilation tool and CMAKE (version 3.10 or higher) is installed. Execute these commands in sequence: ① cd your_project_path ② mkdir build ③ cd build ④ cmake .. ⑤ make

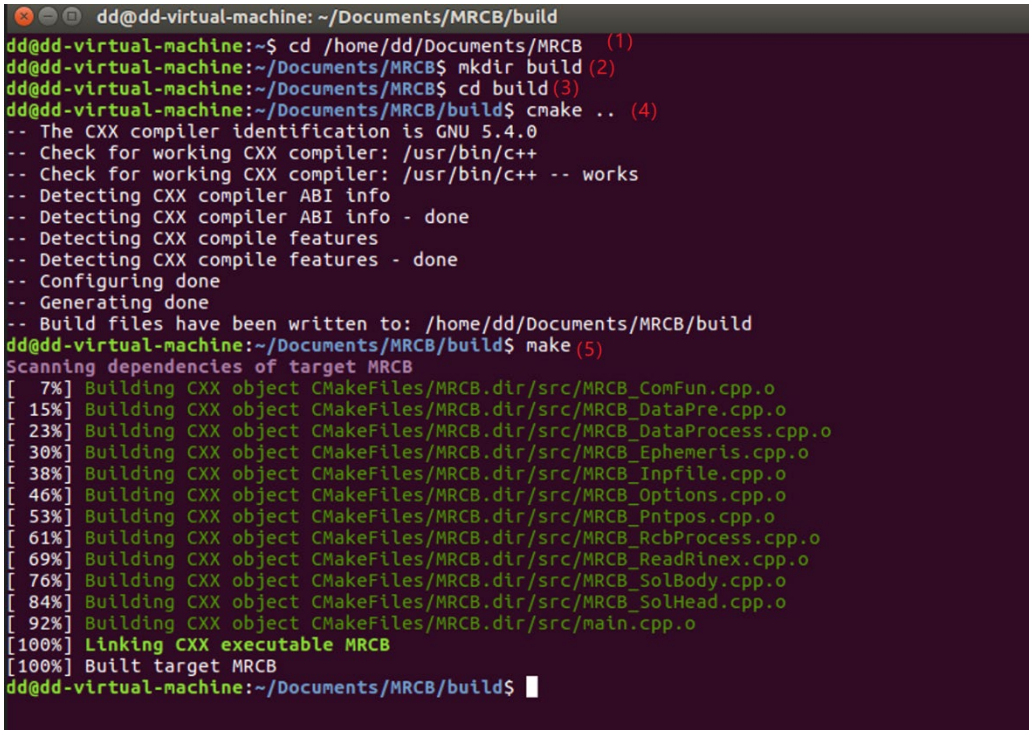


Fig. 11 Finish compile under Linux

6.2 Run

Using Windows as an example, we will introduce the operational process with two examples: the day-by-day process and continuous processing

6.2.1 day-by-day process

（1）Compile code

Reference the **section 6.1.**

（2）Data Preparation

In the **config.dat** file, configure the parameters by setting the input and output paths according to your specific needs and file locations.

```
# Multi-day
# To switch: set Continue = 1, comment out OBS_PATH and NAV_PATH,
#=============================================================
OBS_DIR   =..\Testdata\continuous processing\obs\
NAV_DIR   =..\Testdata\continuous processing\nav\

#=============================================================
# 2. Output
#=============================================================
# OUT_PRO    satellite detail (.pro)
# OUT_PATH   RCB solution (.out); also temperature fit (.tfit) and fit residual (.tres)
# OUT_LOG    rejected-satellite log (.log)
# Directory must already exist (program does not create it).
OUT_PRO   =..\result\
OUT_PATH  =..\result\
OUT_LOG   =..\result\

#=============================================================
# 3. Processing
#=============================================================
# Ephopt     0 broadcast ephemeris
# UseBDS     0 BDS-3, 1 BDS-2, 2 BDS-2+BDS-3 (first two freqs: B1I B3I)
# Frequency  number of frequencies
# Elmask     elevation mask (deg)
# Exclsats   excluded PRNs, separated by spaces
# System     bit mask: GPS 1, GLO 4, GAL 8, QZSS 16, BDS 32
# Continue   0 day-by-day, 1 multi-day
Ephopt     =0
UseBDS     =2
Frequency  =2
Elmask     =10
Exclsats   =C01 C02 C03 C04 C05
System     =45
Continue   =1
#=============================================================
# 4. Station
#=============================================================
StaName    =MRO1
DOY        =2024298
```

Fig. 12 Paths Requiring Configuration

（3）Run code

Recommend running in the command line: ① cd ./build/Debug ② TdRCB.exe

```
C:\Users\Administrator\Desktop\test\build\Debug>TdRCB.exe
temperature: 2880 epochs, 24 h, 30 s, interpolated 0
                                    GLONASS                                 Galileo                                 QZSS                                    BDS
2020 09 25 00 00 00 FLOAT  0.000  0.000  0.000    0.000  0.000    0.000  0.000  0.000  0.000  0.000    0.000  0.000  0.000    0.000  0.000  0.000  0.000  0.000  25  99999.990
2020 09 25 00 00 30 FLOAT -0.232  0.126  0.000   -0.114  0.020    0.090 -0.063  0.000  0.000  0.000    0.000  0.000  0.000    0.043 -0.039  0.000  0.000  0.000  25      0.207
2020 09 25 00 01 00 FLOAT -0.204  0.168  0.000   -0.056 -0.083    0.130 -0.085  0.000  0.000  0.000    0.000  0.000  0.000   -0.291 -0.018  0.000  0.000  0.000  25      0.148
2020 09 25 00 01 30 FLOAT  0.117  0.059  0.000    0.017 -0.038    0.095 -0.051  0.000  0.000  0.000    0.000  0.000  0.000   -0.093 -0.047  0.000  0.000  0.000  25      0.122
2020 09 25 00 02 00 FLOAT -0.174  0.056  0.000   -0.154 -0.109    0.111 -0.007  0.000  0.000  0.000    0.000  0.000  0.000    0.020 -0.009  0.000  0.000  0.000  25      0.107
2020 09 25 00 02 30 FLOAT  0.059 -0.008  0.000   -0.087 -0.026    0.050 -0.066  0.000  0.000  0.000    0.000  0.000  0.000   -0.069 -0.033  0.000  0.000  0.000  25      0.203
2020 09 25 00 03 00 FLOAT  0.248  0.162  0.000   -0.110  0.019    0.086 -0.028  0.000  0.000  0.000    0.000  0.000  0.000   -0.071 -0.002  0.000  0.000  0.000  25      0.152
2020 09 25 00 03 30 FLOAT  0.032  0.104  0.000   -0.009 -0.059    0.032 -0.086  0.000  0.000  0.000    0.000  0.000  0.000   -0.062 -0.011  0.000  0.000  0.000  25      0.218
2020 09 25 00 04 00 FLOAT -0.038  0.080  0.000   -0.182 -0.017   -0.099 -0.022  0.000  0.000  0.000    0.000  0.000  0.000   -0.093 -0.041  0.000  0.000  0.000  25      0.125
2020 09 25 00 04 30 FLOAT -0.120  0.203  0.000   -0.152  0.057    0.126 -0.021  0.000  0.000  0.000    0.000  0.000  0.000   -0.105 -0.066  0.000  0.000  0.000  25      0.125
2020 09 25 00 05 00 FLOAT -0.124  0.159  0.000    0.016  0.068    0.066 -0.032  0.000  0.000  0.000    0.000  0.000  0.000    0.078  0.010  0.000  0.000  0.000  25      0.146
2020 09 25 00 05 30 FLOAT  0.202  0.182  0.000   -0.066  0.063    0.011 -0.041  0.000  0.000  0.000    0.000  0.000  0.000   -0.065 -0.016  0.000  0.000  0.000  25      0.071
2020 09 25 00 06 00 FLOAT  0.226  0.073  0.000   -0.079  0.078   -0.038 -0.000  0.000  0.000  0.000    0.000  0.000  0.000   -0.094  0.062  0.000  0.000  0.000  25      0.197
2020 09 25 00 06 30 FLOAT  0.025  0.001  0.000   -0.019  0.049    0.027 -0.073  0.000  0.000  0.000    0.000  0.000  0.000   -0.025  0.060  0.000  0.000  0.000  25      0.073
2020 09 25 00 07 00 FLOAT -0.218  0.050  0.000   -0.092  0.029    0.030 -0.033  0.000  0.000  0.000    0.000  0.000  0.000    0.032  0.005  0.000  0.000  0.000  25      0.176
2020 09 25 00 07 30 FLOAT  0.071  0.117  0.000   -0.078  0.096    0.088 -0.030  0.000  0.000  0.000    0.000  0.000  0.000   -0.005  0.045  0.000  0.000  0.000  25      0.122
2020 09 25 00 08 00 FLOAT  0.120  0.185  0.000   -0.044  0.082    0.075 -0.039  0.000  0.000  0.000    0.000  0.000  0.000   -0.019 -0.041  0.000  0.000  0.000  25      0.089
2020 09 25 00 08 30 FLOAT -0.003  0.045  0.000    0.022  0.095    0.021 -0.046  0.000  0.000  0.000    0.000  0.000  0.000   -0.075  0.049  0.000  0.000  0.000  25      0.136
2020 09 25 00 09 00 FLOAT  0.119  0.145  0.000   -0.119  0.082    0.011 -0.052  0.000  0.000  0.000    0.000  0.000  0.000   -0.033  0.006  0.000  0.000  0.000  25      0.124
2020 09 25 00 09 30 FLOAT -0.098 -0.019  0.000    0.045  0.041    0.003 -0.039  0.000  0.000  0.000    0.000  0.000  0.000   -0.032  0.015  0.000  0.000  0.000  25      0.217
2020 09 25 00 10 00 FLOAT -0.001  0.143  0.000    0.062  0.055    0.052 -0.056  0.000  0.000  0.000    0.000  0.000  0.000   -0.028 -0.037  0.000  0.000  0.000  25      0.165
```

Fig. 13 Runtime Display

（4）Visualization

Two simple visualization scripts are located in the .\script directory and require MATLAB (version R2020b or higher) to run.

- RCB_day_by_day.m: Visualizes RCBs for a single day. Simply configure the file paths and specify the frequencies and systems to visualize.

RCB_day_by_day.m

```
% Simple script to plot RCBs
clear; close all; clc;
% User settings
filename = 'E:\open_source\result\MRO1_GREJC_2_2024298_RCB.out';
system = 'Galileo';       % Choose: GPS, GLONASS, Galileo, QZSS, BDS-2, BDS-3
numFreq = 2;              % How many frequency bands to plot

% System configuration
bands.GPS = {'P1','P2','P5'}; cols.GPS = 12:14;
bands.GLONASS = {'R1','R2'}; cols.GLONASS = 15:16;
bands.Galileo = {'E1','E5a','E5b','E6','E5'}; cols.Galileo = 17:21;
bands.QZSS = {'P1','P2','P5'}; cols.QZSS = 22:24;
bands.BDS_2 = {'B1I','B3I','B2I'}; cols.BDS_2 = 25:27;
bands.BDS_3 = {'B1I','B3I','B2b','B1C','B2a'}; cols.BDS_3 = 25:29;

% Handle BDS names
if strcmp(system, 'BDS-2'), key = 'BDS_2';
elseif strcmp(system, 'BDS-3'), key = 'BDS_3';
else, key = system;
end

% Read data
data = readmatrix(filename, 'FileType','text', 'CommentStyle','#', 'Delimiter',' ', 'MultipleDelimsAsOne',true, 'NumHeaderLines',0);
hour = data(:,5);

% Get RCB data
```

Fig. 14 Visualizes RCBs for a single day

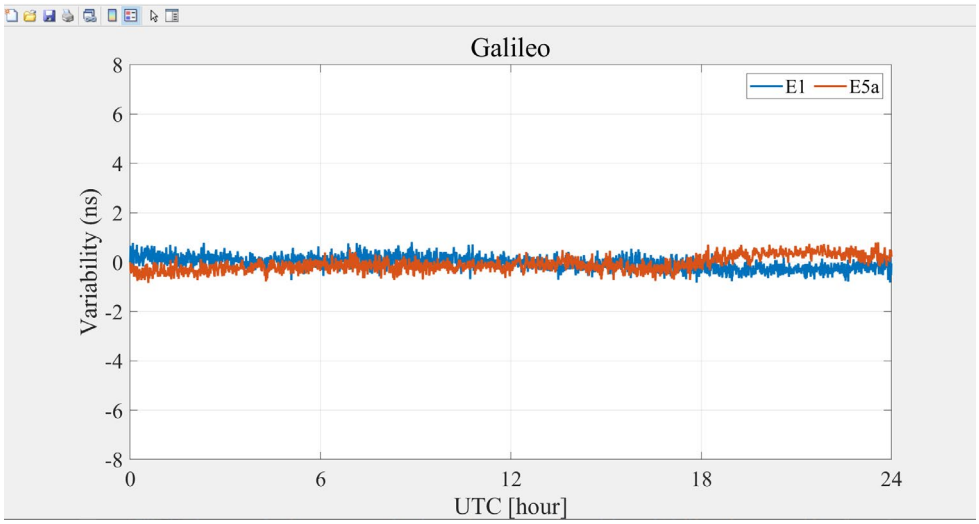


Fig. 15 Visualization Results

6.2.2 continuous processing

（1）Compile code

Reference the **section 6.1.**

（2）Data Preparation

In the **config.dat** file, configure the parameters by setting the input and output paths according to your specific needs and file locations. When use **continuous processing,** the following parameters do not need to be set and can be left as default: OBS_PATH, NAV_PATH, DOY. Configure the following input and output parameters:

```
# Multi-day
# To switch: set Continue = 1, comment out OBS_PATH and NAV_PATH,
#-------------------------------------------------------------------
OBS_DIR   =..\Testdata\continuous processing\obs\
NAV_DIR   =..\Testdata\continuous processing\nav\

#===================================================================
# 2. Output
#===================================================================
# OUT_PRO    satellite detail (.pro)
# OUT_PATH   RCB solution (.out); also temperature fit (.tfit) and fit residual (.tres)
# OUT_LOG    rejected-satellite log (.log)
# Directory must already exist (program does not create it).
OUT_PRO   =..\result\
OUT_PATH  =..\result\
OUT_LOG   =..\result\

#===================================================================
# 3. Processing
#===================================================================
# Ephopt     0 broadcast ephemeris
# UseBDS     0 BDS-3, 1 BDS-2, 2 BDS-2+BDS-3 (first two freqs: B1I B3I)
# Frequency  number of frequencies
# Elmask     elevation mask (deg)
# Exclsats   excluded PRNs, separated by spaces
# System     bit mask: GPS 1, GLO 4, GAL 8, QZSS 16, BDS 32
# Continue   0 day-by-day, 1 multi-day
Ephopt     =0
UseBDS     =2
Frequency  =2
Elmask     =10
Exclsats   =C01 C02 C03 C04 C05
System     =45
Continue   =1
#===================================================================
# 4. Station
#===================================================================
StaName    =MRO1
DOY        =2024298
```

Fig. 16 Paths Requiring Configuration

（3）Run code

Recommend running in the command line.

（4）Visualization

- RCB_Continues.m: Visualizes RCBs for continues day. Simply configure the file paths and specify the frequencies and systems to visualize.

RCB_Continues.m

```
% Simple script to plot RCBs
clear; close all; clc;
% User settings
filename = 'E:\open_source\result\MRO1_GREJC_2_Continues_RCB.out';
system = 'Galileo';      % Choose: GPS, GLONASS, Galileo, QZSS, BDS-2, BDS-3
numFreq = 2;             % How many frequency bands to plot

% System configuration
bands.GPS = {'P1','P2','P5'}; cols.GPS = 12:14;
bands.GLONASS = {'R1','R2'}; cols.GLONASS = 15:16;
bands.Galileo = {'E1','E5a','E5b','E6','E5'}; cols.Galileo = 17:21;
bands.QZSS = {'P1','P2','P5'}; cols.QZSS = 22:24;
bands.BDS_2 = {'B1I','B3I','B2I'}; cols.BDS_2 = 25:27;
bands.BDS_3 = {'B1I','B3I','B2b','B1C','B2a'}; cols.BDS_3 = 25:29;

% Handle BDS names
if strcmp(system, 'BDS-2'), key = 'BDS_2';
elseif strcmp(system, 'BDS-3'), key = 'BDS_3';
else, key = system;
end

% Read data
data = readmatrix(filename, 'FileType','text', 'CommentStyle','#', 'Delimiter',' ', 'MultipleDelimsAsOne',true, 'NumHeaderLines',0);
hour = data(:,4);
```

Fig. 17 Visualizes RCBs for continues day

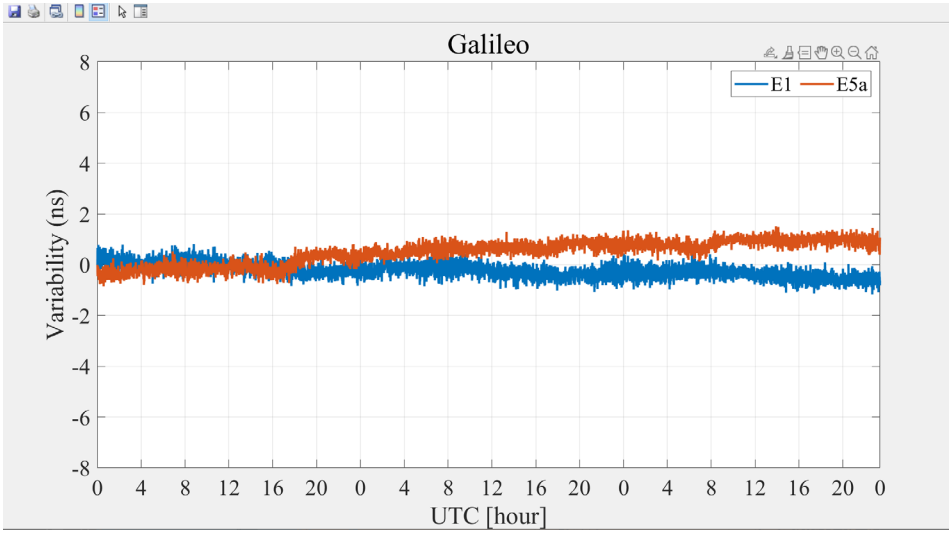


Fig. 18 Visualization Results

6.2.3 Linux Run

If you want run on the Linux system.

Execute these commands in sequence: ./TdRCB

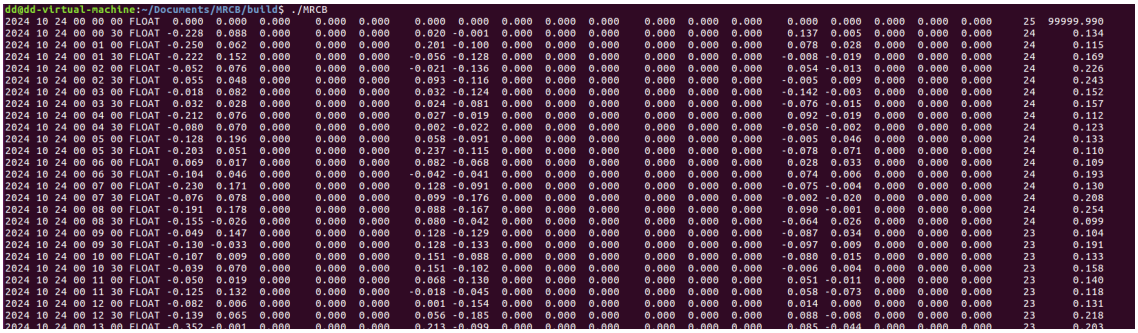


Fig. 19 Run on Linux

7 Output file format description

7.1 Result File

Name: XXXX_GEC_2_YYYYDOY_RCB.out

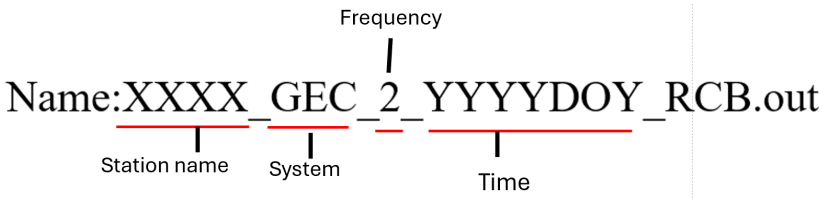
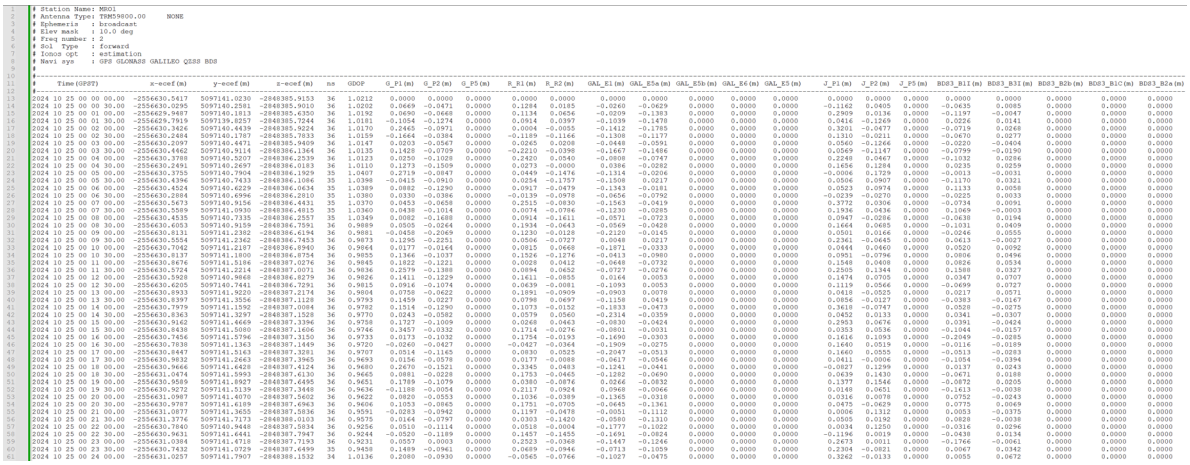


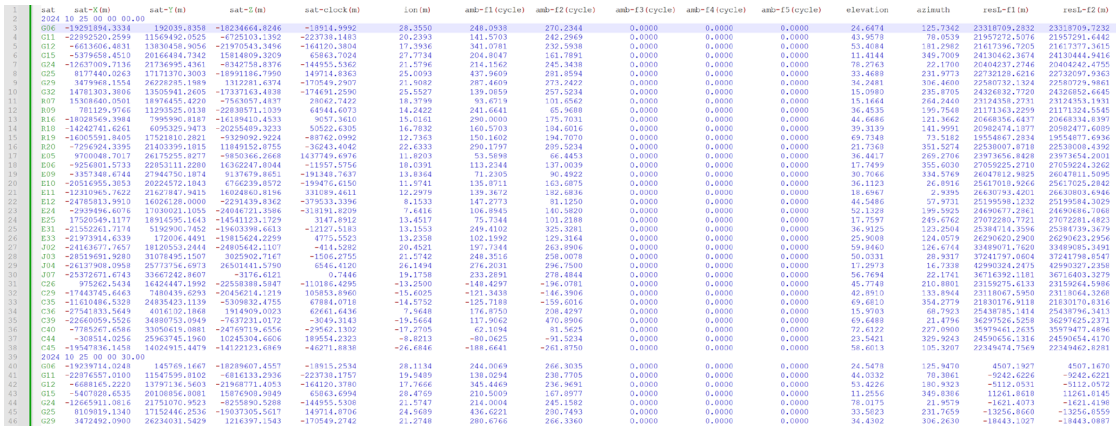
Fig. 20 Structure of output file name

Table 3 Result output file format description

Items		Value Type	Range
Epoch Time (GPST)	Year	char	1-4
	Month	char	6-7
	Day	char	9-10
	Hour	char	12-13
	Minute	char	15-16
	Second	char	18-22
Receiver Position (SPP)	X (m)	double	25-37
	Y (m)	double	41-52
	Z (m)	double	55-67
Number of valid satellites		int	71-72
GDOP		double	76-81
GPS Time-varying RCB	P1	double	86-91
	P2	double	95-100
	P5	double	104-109
GLONASS Time-varying RCB	R1	double	115-120
	R2	double	124-130
Galileo Time-varying RCB	E1	double	136-141
	E5a	double	146-151
	E5b	double	157-162
	E6	double	167-172
	E5	double	177-182
QZSS Time-varying RCB	P1	double	190-195
	P2	double	199-204
	P5	double	208-213
BDS-3 Time-varying RCB	B1I	double	218-223
	B3I	double	230-235
	B2b	double	242-248
	B1C	double	255-260
	B2a	double	266-271
BDS-2 Time-varying RCB	B1I	double	218-223
	B3I	double	230-235
	B2I	double	242-248



```
# Station Name: MS01
# Antenna Type: TRM59800.00     NONE
# Ephemeris   : broadcast
# Elev mask   : 10.0 deg
# Freq number : 2
# Sol Type    : forward
# Ionos opt   : estimation
# Navi sys    : GPS GLONASS GALILEO QZSS BDS
#
#-------------------
```



sat	sat-X(m)	sat-Y(m)	sat-Z(m)	sat-clock(m)	ion(m)	amb-f1(cycle)	amb-f2(cycle)	amb-f3(cycle)	amb-f4(cycle)	amb-f5(cycle)	elevation	azimuth	resL-f1(m)	resL-f2(m)
2024 10 25 00 00 00.00														
G06	-19251894.3334	192039.8358	-18234664.8246	-18914.9992	28.3550	248.0938	270.2344	0.0000	0.0000	0.0000	24.6474	125.7342	23318709.2832	23318709.7232
G11	-22392520.2599	11569492.0525	-6725103.1392	-223738.1483	20.2393	141.5703	242.2969	0.0000	0.0000	0.0000	43.9578	78.0539	21957272.5076	21957291.6442
G12	-6613606.4831	13830458.9056	-21970543.3496	-164120.3804	17.3936	341.0781	232.5938	0.0000	0.0000	0.0000	53.4084	181.2982	21617396.7205	21617377.3615
G15	-5379658.4510	20166404.7342	15814809.3209	65863.7024	27.7734	204.8047	161.7891	0.0000	0.0000	0.0000	11.4144	349.7009	24130462.3674	24130444.9416
G24	-12637009.7136	21736995.4361	-8342758.8376	-144955.5362	21.5796	214.1562	245.3438	0.0000	0.0000	0.0000	78.2763	22.1700	20404237.2746	20404242.4755
G25	8177440.0263	17171376.3003	-18991186.7950	149714.8363	25.0093	437.9609	291.8594	0.0000	0.0000	0.0000	33.4688	231.9773	22732128.6216	22732097.9363
G29	3479968.1554	26228285.1989	1312281.6374	-170549.2907	21.9082	287.4609	273.2422	0.0000	0.0000	0.0000	34.2481	306.4600	22580732.1324	22580729.9861
G32	14781303.3806	13505941.2605	-17337163.4838	-174691.2590	25.5527	139.0859	257.5234	0.0000	0.0000	0.0000	15.0980	235.8705	24326832.7720	24326852.6645
R07	15308640.0501	18976455.4220	-7563057.4837	28062.7422	18.3799	93.6719	101.6562	0.0000	0.0000	0.0000	15.1664	264.2460	23124358.2731	23124353.1931
R09	781129.9766	11293525.0138	-22830571.1039	64944.6073	14.2422	241.6641	65.9688	0.0000	0.0000	0.0000	36.4535	195.7548	21171363.2259	21171324.5545
R16	-18028569.3984	7995996.8187	-16198410.4533	9057.3410	15.0161	290.0000	175.7031	0.0000	0.0000	0.0000	44.6686	121.3662	20668356.4837	20668334.8397
R18	-14242741.6261	6095325.9473	-20255489.3233	50522.6305	16.7832	160.5703	194.6016	0.0000	0.0000	0.0000	39.3139	141.9991	20982474.1877	20982477.6089
R19	-16005591.8405	17521810.2821	-9329092.9224	-88762.0992	12.7363	150.1602	194.7070	0.0000	0.0000	0.0000	69.7348	73.5182	19554867.2834	19554877.6936
R20	-7296924.3395	21403396.1815	11849152.8755	-36243.4042	22.6333	290.1797	295.5234	0.0000	0.0000	0.0000	21.7360	351.5274	22538007.8718	22538008.4392
E05	9700948.7017	26175255.9277	-9850366.2668	1437749.6976	11.8203	53.5898	66.4453	0.0000	0.0000	0.0000	36.4417	269.2706	23973656.8428	23973654.2001
E06	-9256001.5733	22853111.2280	16362247.8044	-11957.5756	10.0391	113.2344	137.0039	0.0000	0.0000	0.0000	17.7499	355.6030	27059225.2710	27059224.3262
E09	-3357348.6744	27942790.1874	9137679.8651	-191348.7637	13.8364	71.2305	90.4922	0.0000	0.0000	0.0000	30.7066	334.5769	26047811.9025	26047811.5095
E10	-20516955.3653	20224572.1843	6766239.8572	-199476.6150	11.5741	135.8711	163.6875	0.0000	0.0000	0.0000	36.1123	26.8916	25617018.9266	25617025.2842
E11	-12310965.7622	21627847.9415	16024860.8196	331089.4611	12.2979	139.3672	182.6836	0.0000	0.0000	0.0000	18.6967	2.9395	26630793.4201	26630803.6946
E12	-24785813.9910	16026128.0000	-2291439.8362	-379533.3396	8.1533	147.2773	81.1250	0.0000	0.0000	0.0000	44.5486	57.9731	25195584.1232	25195584.3025
E24	-2939496.6076	17030021.1055	-24046721.3586	-318191.8209	7.6416	106.8945	140.5820	0.0000	0.0000	0.0000	52.1328	199.5925	24690677.2861	24690686.7068
E25	17520549.1177	18914595.1643	-14541123.1729	3147.8912	13.4517	75.7344	101.2188	0.0000	0.0000	0.0000	17.7597	245.6762	27072280.7721	27072281.4823
E31	-21552261.7174	5192906.7452	-11603298.6613	-121127.5193	13.1553	249.4102	325.3281	0.0000	0.0000	0.0000	36.9125	123.2504	25384739.3596	25384739.3679
E33	-21973914.6339	172006.4491	-19815624.2259	4775.5523	13.2358	102.1592	129.3164	0.0000	0.0000	0.0000	25.9008	124.0579	26290620.2900	26290623.2956
J02	-24143677.7657	18120553.2444	-24805442.1107	-414.5282	20.4521	197.7344	263.8906	0.0000	0.0000	0.0000	59.8460	126.6744	33489071.7620	33489085.3491
J03	-28519691.9280	31078495.1507	3025902.7167	-15506.2755	21.5742	248.3516	258.0078	0.0000	0.0000	0.0000	50.0331	28.9317	37241797.0604	37241798.8547
J04	-26137908.0958	25773756.6973	26501441.5790	6546.4120	26.1494	276.2031	296.7500	0.0000	0.0000	0.0000	17.2973	16.7338	42990324.2475	42990327.2358
J07	-25372671.6743	33667242.9607	-3176.6121	0.7446	19.1758	233.2891	278.4844	0.0000	0.0000	0.0000	56.7694	22.1741	36716392.1181	36716403.3279
C26	975262.5434	16424447.1992	-22558882.5947	-110186.4295	-13.2500	-148.4297	-196.0781	0.0000	0.0000	0.0000	45.7748	216.8801	23159275.6133	23159264.5906
C29	-17443745.6463	7480439.6293	-20456214.1219	105853.8960	-15.6025	-121.3438	-146.3906	0.0000	0.0000	0.0000	42.8910	133.8944	23118067.5950	23118064.3268
C35	-11610486.5328	24035423.1139	-5309832.4755	67884.0718	-14.5752	-125.7188	-159.6016	0.0000	0.0000	0.0000	69.6810	354.2779	21830176.9118	21830170.8316
C36	-27541833.5649	4016102.1868	1914909.0023	62661.6436	7.9648	176.8750	208.4297	0.0000	0.0000	0.0000	15.9703	68.7923	25438785.1414	25438796.3413
C39	-22660059.5526	34880753.0949	-7637231.0172	-3049.3143	-19.5664	117.9062	470.8906	0.0000	0.0000	0.0000	69.6488	21.4796	36297526.5258	36297625.2371
C40	-7785367.4586	33050619.0881	-24769719.6356	-29562.1302	-17.2705	62.1094	91.5625	0.0000	0.0000	0.0000	72.6122	227.0900	35979441.2635	35979477.4896
C44	-308514.0256	25963745.1960	10245304.6606	189554.2323	-8.8213	-80.0625	-91.5234	0.0000	0.0000	0.0000	23.5421	325.9243	24590656.1316	24590654.4170
C45	-19547836.1458	14024915.4479	-14122123.6869	-46271.8838	-26.6846	-188.6641	-261.8750	0.0000	0.0000	0.0000	58.6013	105.3207	22349474.7569	22349462.8281
2024 10 25 00 00 30.00														
G06	-19239714.0248	145769.1667	-18289607.4557	-18915.2534	28.1134	244.0069	266.3035	0.0000	0.0000	0.0000	24.5478	125.9470	4507.1927	4507.1670
G11	-22376557.0100	11547599.0102	-6816133.2936	-223738.1757	19.5489	138.0294	238.7705	0.0000	0.0000	0.0000	44.0332	78.3861	-9242.6226	-9242.6221
G12	-6608165.2220	13797136.5603	-21968771.4353	-164120.3780	17.7666	345.4469	236.9691	0.0000	0.0000	0.0000	53.4226	180.9323	-5112.0531	-5112.0572
G15	-5407028.6535	20108856.8081	15876900.9949	65863.6994	28.4769	210.5009	167.8977	0.0000	0.0000	0.0000	11.2556	345.8386	11261.8618	11261.8145
G24	-12665911.0816	21751070.9523	-8255890.5288	-144955.5308	21.5747	214.0004	245.1582	0.0000	0.0000	0.0000	78.0175	21.9579	-1621.4073	-1621.4198
G25	8169819.1340	17152446.2536	-19037305.5617	149714.8706	24.5609	436.6221	290.7493	0.0000	0.0000	0.0000	33.5823	231.7659	-13256.8660	-13256.8559
G29	3472492.0900	26234031.5429	1216397.1543	-170549.2742	21.2748	280.6766	266.3360	0.0000	0.0000	0.0000	34.4302	306.2630	-18443.1027	-18443.0887

Fig. 22 Screenshot of Process Information file

7.3 Log file

Name: XXXX_GEC_2_YYYYDOY_Sat.log

The log file mainly contains the record of satellite eliminations.

```
2024/10/25 00:00:00.00
Deleter Sat for lack observations, need freq 3:G12(Frq:2)
Deleter Sat for lack observations, need freq 3:G15(Frq:2)
Deleter Sat for lower 10.0 degree:G19(Ele:8.6)
Deleter Sat for lower 10.0 degree:G20(Ele:2.8)
Deleter Sat for lack observations, need freq 3:G29(Frq:2)
Deleter Sat for lower 10.0 degree:E03(Ele:8.8)
Deleter Sat for lower 10.0 degree:C21(Ele:6.6)
Deleter Sat for lower 10.0 degree:C42(Ele:5.1)
Deleter Sat for lower 10.0 degree:C50(Ele:7.4)
2024/10/25 00:00:30.00
Deleter Sat for lack observations, need freq 3:G12(Frq:2)
Deleter Sat for lack observations, need freq 3:G15(Frq:2)
Deleter Sat for lower 10.0 degree:G19(Ele:8.5)
Deleter Sat for lower 10.0 degree:G20(Ele:2.9)
Deleter Sat for lack observations, need freq 3:G29(Frq:2)
Deleter Sat for lower 10.0 degree:E03(Ele:8.9)
Deleter Sat for lower 10.0 degree:C21(Ele:6.5)
Deleter Sat for lower 10.0 degree:C42(Ele:5.1)
Deleter Sat for lower 10.0 degree:C50(Ele:7.3)
2024/10/25 00:01:00.00
Deleter Sat for lack observations, need freq 3:G12(Frq:2)
Deleter Sat for lack observations, need freq 3:G15(Frq:2)
Deleter Sat for lower 10.0 degree:G19(Ele:8.3)
Deleter Sat for lack observations, need freq 3:G29(Frq:2)
Deleter Sat for lower 10.0 degree:E03(Ele:9.0)
Deleter Sat for lower 10.0 degree:C21(Ele:6.4)
Deleter Sat for lower 10.0 degree:C42(Ele:5.0)
Deleter Sat for lower 10.0 degree:C50(Ele:7.2)
2024/10/25 00:01:30.00
Deleter Sat for lack observations, need freq 3:G12(Frq:2)
Deleter Sat for lack observations, need freq 3:G15(Frq:2)
Deleter Sat for lower 10.0 degree:G19(Ele:8.1)
Deleter Sat for lack observations, need freq 3:G29(Frq:2)
Deleter Sat for lower 10.0 degree:E03(Ele:9.0)
Deleter Sat for lower 10.0 degree:C21(Ele:6.3)
Deleter Sat for lower 10.0 degree:C42(Ele:4.9)
Deleter Sat for lower 10.0 degree:C50(Ele:7.1)
```

Fig. 23 Screenshot of Log file

7.4 tfit File

This file provides the quadratic fitting models relating temperature (T) (°C) to receiver code bias (RCB) (m) for each GNSS constellation and frequency.

Field	Description
System	GNSS constellation: GPS, GLONASS, Galileo, QZSS, or BDS.
Band	Frequency identifier, such as G_P1 or BDS_B1I.
N	Number of epochs included in the fitting process.
a	Quadratic coefficient（m/℃²）
b	Linear coefficient（m/℃）
c	Constant term（m）
R2	Coefficient of determination R^2, indicating the goodness of fit.
Equation	$RCB(T) = aT^2 + bT + c$

```
# Station Name: LONA
# Ephemeris   : broadcast
# Elev mask   : 10.0 deg
# Freq number : 2
# Ionos opt   : estimation
# Navi sys    : GPS GLONASS GALILEO BDS
# Temperature-RCB quadratic fit
# Model   : RCB(m) = a * T(degC)^2 + b * T(degC) + c
# Epochs  : float RCB solutions that have a temperature
#
System      Band          N        a(m/degC^2)          b(m/degC)              c(m)        R2  Equation
GPS         G_P1       2879         1.991001e-02        -5.225417e-01       2.789649e+00   0.9583  RCB = 1.991001e-02*T^2 -5.225417e-01*T +2.789649e+00
GPS         G_P2       2879         2.424611e-02        -6.898060e-01       4.695192e+00   0.9728  RCB = 2.424611e-02*T^2 -6.898060e-01*T +4.695192e+00
GLONASS     R_R1       2879         2.560509e-02        -7.592429e-01       5.548247e+00   0.9721  RCB = 2.560509e-02*T^2 -7.592429e-01*T +5.548247e+00
GLONASS     R_R2       2879         2.418823e-02        -7.158628e-01       5.189837e+00   0.9742  RCB = 2.418823e-02*T^2 -7.158628e-01*T +5.189837e+00
Galileo     GAL_E1     2879         2.420992e-02        -6.628667e-01       4.032129e+00   0.9689  RCB = 2.420992e-02*T^2 -6.628667e-01*T +4.032129e+00
Galileo     GAL_E5a    2879         2.645982e-02        -7.569228e-01       4.995177e+00   0.9727  RCB = 2.645982e-02*T^2 -7.569228e-01*T +4.995177e+00
BDS         BDS_B1I    2879         2.966523e-02        -8.347675e-01       5.051123e+00   0.9693  RCB = 2.966523e-02*T^2 -8.347675e-01*T +5.051123e+00
BDS         BDS_B3I    2879         3.150326e-02        -8.975639e-01       5.822207e+00   0.9733  RCB = 3.150326e-02*T^2 -8.975639e-01*T +5.822207e+00
```

Fig. 24 Screenshot of .tfit file

7.5 tres File

This file contains the fitting residuals at each epoch.

Residual definition: $residual = RCB_{solved} - RCB_{model}$

Units: RCB, RCBfit, and residual are expressed in meters (m), while temperature is expressed in degrees Celsius (°C).

Field	Description
Time(GPST)	Epoch time in GPS Time (GPST), represented by year, month, day, hour, minute, and second.
System	GNSS constellation name.
Band	Frequency identifier.
T(degC)	Temperature at the corresponding epoch (°C).
RCB(m)	Estimated receiver code bias at the corresponding epoch (m).
RCBfit(m)	RCB predicted by the fitted temperature model (m).
residual(m)	Fitting residual, calculated as the estimated RCB minus the model-predicted RCB (m).

```
# Temperature-RCB fit residual
# residual = RCB_solved - RCB_model,  RCB_model = a*T^2 + b*T + c
# Only estimated system/bands are listed
# GPS         G_P1          RCB = 1.991001e-02*T^2 -5.225417e-01*T +2.789649e+00   (N=2879)
# GPS         G_P2          RCB = 2.424611e-02*T^2 -6.898060e-01*T +4.695192e+00   (N=2879)
# GLONASS     R_R1          RCB = 2.560509e-02*T^2 -7.592429e-01*T +5.548247e+00   (N=2879)
# GLONASS     R_R2          RCB = 2.418823e-02*T^2 -7.158628e-01*T +5.189837e+00   (N=2879)
# Galileo     GAL_E1        RCB = 2.420992e-02*T^2 -6.628667e-01*T +4.032129e+00   (N=2879)
# Galileo     GAL_E5a       RCB = 2.645982e-02*T^2 -7.569228e-01*T +4.995177e+00   (N=2879)
# BDS         BDS_B1I       RCB = 2.966523e-02*T^2 -8.347675e-01*T +5.051123e+00   (N=2879)
# BDS         BDS_B3I       RCB = 3.150326e-02*T^2 -8.975639e-01*T +5.822207e+00   (N=2879)
#
Time(GPST)          System     Band             T(degC)        RCB(m)      RCBfit(m)     residual(m)
2020 09 25 00 00 30.0 GPS        G_P1               16.4000     -0.231502      -0.425039       0.193537
2020 09 25 00 01 00.0 GPS        G_P1               16.3000     -0.204316      -0.437891       0.233575
2020 09 25 00 01 30.0 GPS        G_P1               16.3000      0.116517      -0.437891       0.554408
2020 09 25 00 02 00.0 GPS        G_P1               16.3000     -0.174435      -0.437891       0.263456
2020 09 25 00 02 30.0 GPS        G_P1               16.4000      0.059301      -0.425039       0.484340
2020 09 25 00 03 00.0 GPS        G_P1               16.3000      0.247854      -0.437891       0.685745
2020 09 25 00 03 30.0 GPS        G_P1               16.4000      0.032347      -0.425039       0.457386
2020 09 25 00 04 00.0 GPS        G_P1               16.3000     -0.038058      -0.437891       0.399833
2020 09 25 00 04 30.0 GPS        G_P1               16.3000     -0.120202      -0.437891       0.317689
2020 09 25 00 05 00.0 GPS        G_P1               16.3000     -0.123668      -0.437891       0.314222
```

Fig. 25 Screenshot of . tres file

7.6 Execution Interface

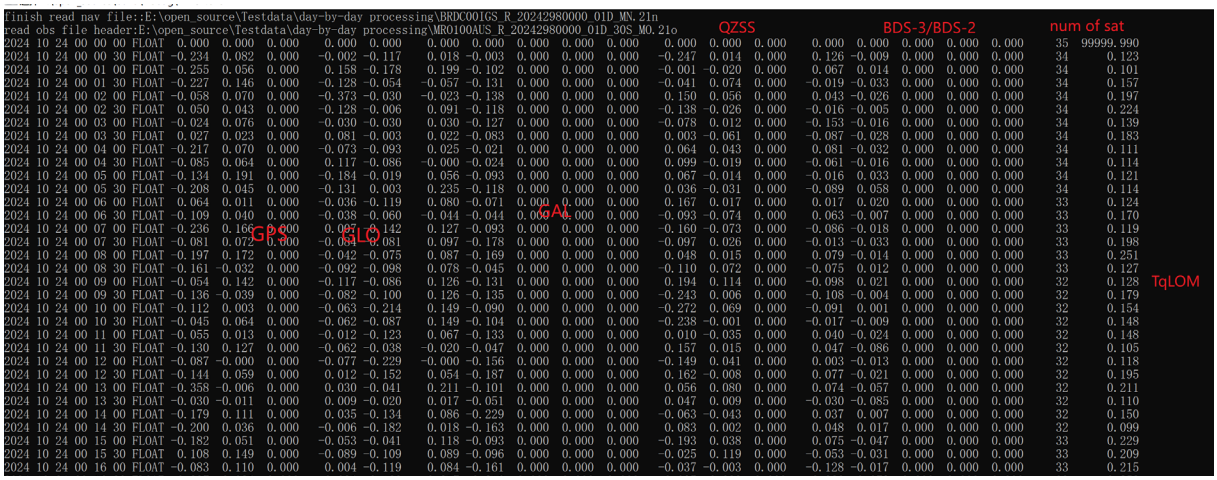


Fig. 27 Runtime Display RCBs

Appendix

(1) CDMA RCB Function Model

We commence with the undifferenced and uncombined GNSS observation equations, which are written as (Teunissen and Montenbruck, 2017)

$$\begin{cases} E[P_{r,f}^s(i)] = \rho_r^s(i) + \mu_f \cdot I_{r,1}^s(i) + b_{r,f} - b_f^s \\ E[L_{r,f}^s(i)] = \rho_r^s(i) - \mu_f \cdot I_{r,1}^s(i) + \lambda_f \alpha_{r,f}^s \end{cases} \tag{1}$$

where $E(.)$ denotes the expectation operator. $P_{r,f}^s$ and $L_{r,f}^s$ represent observed-minus-calculated code and phase observations of a satellite s tracked by a receiver r on the frequency f at the epoch i, respectively. The frequency-independent parameters in this, including the range, receiver clock offsets, satellite clock offsets, and tropospheric delays, were combined into a single parameter $\rho_r^s(i)$. the first-order slant ionospheric delay $I_{r,1}^s(i)$ on the first frequency linked to other frequencies by the coefficient $\mu_f = \lambda_f^2/\lambda_1^2$ with λ_f being the wavelength, the RCB $b_{r,f}$ and its satellite counterpart b_f^s, and the float ambiguity $\alpha_{r,f}^s = N_{r,f}^s + \delta_{r,f} - \delta_f^s$ containing the receiver phase bias (RPB) $\delta_{r,f}$, the satellite phase bias δ_f^s, and the integer ambiguity $N_{r,f}^s$. We consider the parameters with an epoch index i as time-varying quantities and assume the parameters without the epoch index to be time constant.

Departing from the time-constant assumption of the RCB, we consider it a time-varying parameter and rewrite Eq. (1) as

$$\begin{cases} E[P_{r,f}^s(i)] = \rho_r^s(i) + \mu_f \cdot I_{r,1}^s(i) + b_{r,f}(1) + \tilde{b}_{r,f}(i) - b_f^s \\ E[L_{r,f}^s(i)] = \rho_r^s(i) - \mu_f \cdot I_{r,1}^s(i) + \lambda_f \alpha_{r,f}^s \end{cases} \tag{2}$$

where

$$\tilde{b}_{r,f}(i) = b_{r,f}(i) - b_{r,f}(1) \tag{3}$$

denotes the time-differenced receiver bias between epoch i and the first epoch, but parameters in Eq. (2) are not uniquely estimable due to the rank deficient problem. For this issue, we turn to the S-system theory (Baarda, 1973; Teunissen, 1985), which constraints a minimum set of parameters and formulates a full-rank model estimating the linear functionals of the original parameters. To make it clear, we first decompose the receiver and satellite code biases on the first two frequencies as

$$(.)_{r,f} = (.)_{r,IF} + \mu_f (.)_{r,GF} \tag{4}$$

where the geometry-free combination of a frequency-dependent quantity is defined as

$$(.)_{GF} = \frac{1}{\mu_2 - \mu_1}[(.)_2 - (.)_1] \tag{5}$$

Inserting Eq. (4) into the code observations equation in Eq. (2) yields

$$\begin{cases} E[P_{r,f}^s(i)] = \rho_r^s(i) + \mu_f I_{r,1}^s(i) + b_{r,IF}(1) + \mu_f b_{r,IF}(1) - \mu_f b_{r,GF}^s + \tilde{b}_{r,f}(i) + \bar{b}_{r,f>2}^s \\ E[L_{r,f}^s(i)] = \rho_r^s(i) - \mu_f I_{r,1}^s(i) + \lambda_f \alpha_{r,f}^s \end{cases} \tag{6}$$

where

$$\bar{b}_{r,f>2}^s = [b_{r,f}(1) - b_{r,IF}(1) - u_f \cdot b_{r,GF}(1)] - (b_{,f}^s - u_f \cdot b_{,GF}^s)$$

denotes the combination of receiver and satellite code biases on the third frequency and above.The rank deficiency elimination for Eq. (6) becomes straightforward, as we can lump the parameters that have common coefficients:

$$E[P_{r,f}^s(i)] = \bar{\rho}_r^s(i) + \mu_f \bar{I}_{r,1}^s(i) + \tilde{b}_{r,f}(i) + \bar{b}_{r,f>2}^s \tag{7}$$

where

$$\begin{cases} \bar{\rho}_r^s(i) = \rho_r^s(i) + b_{r,IF}(1) \\ \bar{I}_{r,1}^s(i) = I_{r,1}^s(i) + b_{r,GF}(1) - b_{,GF}^s \end{cases} \tag{8}$$

denote the frequency-independent parameter and ionospheric delays, respectively. When dealing with the phase observations in Eq. (2), we first replace $\rho_r^s(i)$ and $I_{r,1}^s(i)$ with $\bar{\rho}_r^s(i)$ and $\bar{I}_{r,1}^s(i)$, resulting in additional code biases. By lumping these biases with ambiguities, we obtain the full-rank observation equations as

$$\begin{cases} E[P_{r,f}^s(i)] = \bar{\rho}_r^s(i) + \mu_f \bar{I}_{r,1}^s(i) + \tilde{b}_{r,f}(i) + \bar{b}_{r,f>2}^s \\ E[L_{r,f}^s(i)] = \bar{\rho}_r^s(i) - \mu_f \bar{I}_{r,1}^s(i) + \lambda_f \bar{\alpha}_{r,f}^s \end{cases} \tag{9}$$

where

$$\bar{\alpha}_{r,f}^s = \alpha_{r,f}^s + \left[b_{,IF}^s - b_{r,IF}(1)\right]/\lambda_f - \mu_f \cdot \left[b_{,GF}^s - b_{r,GF}(1)\right]/\lambda_f \tag{10}$$

denote the estimable parameters contain the RCB at the first epoch.

(2) FDMA RCB Function Model

GLONASS raw code and carrier phase observation equations as (Teunissen, 2019)

$$\begin{cases} E[P_{r,f}^s(i)] = \rho_r^s(i) + \mu_f^s \cdot I_{r,1}^s(i) + b_{r,f} - b_f^s + k^s \Delta b_{r,f} \\ E[L_{r,f}^s(i)] = \rho_r^s(i) - \mu_f^s \cdot I_{r,1}^s(i) + \lambda_f^s(\alpha_{r,f}^s + k^s \Delta \delta_{r,f}) \end{cases} \tag{11}$$

different from CDMA observation equations, there exist receiver code and phase biases that are linear functions of the satellite frequency channel number, $\Delta b_{r,f}$ and $\Delta \delta_{r,f}$, respectively. The meanings of the remaining parameters are the same as in Eq. (1).

In particular, GLONASS satellites have different wavelengths, expressed as λ_f^s. Furthermore, the coefficients of the ionospheric parameters μ_f^s are also satellite-dependent. To simplify the coefficients of the ionospheric parameters, they are defined based on the frequency of each GLONASS R1 and R2 satellite as follows:

$$F_f^s = F_f^0 + k^s \Delta F_f \tag{12}$$

where F_f^0 is the center frequency, ΔF_f is the frequency offset, $\Delta F_1 = 9/16$, $\Delta F_2 = 7/16$, k^s is the frequency channel number of satellite s, and they satisfy $|k^s| \leq 13$, $\Delta F_1/F_1^0 = 1/2848$, $\Delta F_2/F_2^0 = 1/2848$. Therefore, the wavelength λ_f^s can be expressed as:

$$\lambda_f^s = \frac{c}{F_f^0 + k^s \Delta F_f} = \frac{\lambda_f^0}{1 + \frac{k^s \Delta F_f}{F_f^0}} = \frac{2848\lambda_f^0}{2848 + k^s} \tag{13}$$

where $\lambda_f^0 = c/F_f^0$ is the wavelength corresponding to the center frequency. According to Equation (13), the coefficients of the ionospheric parameters can be written as:

$$\mu_f^s = \left(\frac{\lambda_f^s}{\lambda_1^s}\right)^2 = \left(\frac{2848\lambda_f^0}{2848 + k^s} \cdot \frac{2848 + k^s}{2848\lambda_1^0}\right)^2 = \mu_f \tag{14}$$

Therefore, Equation (11) can be simplified as:

$$\begin{cases} E[P_{r,f}^s(i)] = \rho_r^s(i) + \mu_f \cdot I_{r,1}^s(i) + b_{r,f} - b_f^s + k^s \Delta b_{r,f} \\ E[L_{r,f}^s(i)] = \rho_r^s(i) - \mu_f \cdot I_{r,1}^s(i) + \lambda_f^s(\alpha_{r,f}^s + k^s \Delta\delta_{r,f}) \end{cases} \tag{15}$$

Departing from the time-constant assumption of the RCB, we consider it a time-varying parameter and rewrite Eq. (14) as:

$$\begin{cases} E[P_{r,f}^s(i)] = \rho_r^s(i) + \mu_f \cdot I_{r,1}^s(i) + b_{r,f}(1) + \tilde{b}_{r,f}(i) - b_f^s + k^s \Delta b_{r,f}(1) + k^s \Delta\bar{b}_{r,f}(i) \\ E[L_{r,f}^s(i)] = \rho_r^s(i) - \mu_f \cdot I_{r,1}^s(i) + \lambda_f^s(\alpha_{r,f}^s + k^s \Delta\delta_{r,f}) \end{cases} \tag{16}$$

where

$$\begin{cases} \tilde{b}_{r,f}(i) = b_{r,f}(i) - b_{r,f}(1) \\ \Delta\bar{b}_{r,f}(i) = \Delta b_{r,f}(i) - \Delta b_{r,f}(1) \end{cases} \tag{17}$$

Note that, apart from the IFBs, the structure of the GLONASS equations is entirely consistent with that of CDMA systems. Therefore, following the rank deficiency elimination procedure used for CDMA, the following equations are obtained:

$$\begin{cases} E[P_{r,f}^s(i)] = \bar{\rho}_r^s(i) + \mu_f \bar{I}_{r,1}^s(i) + \tilde{b}_{r,f}(i) + k^s \Delta\bar{b}_{r,f}(i) \\ E[L_{r,f}^s(i)] = \bar{\rho}_r^s(i) - \mu_f \bar{I}_{r,1}^s(i) + \lambda_f^s(\bar{\alpha}_{r,f}^s + k^s \Delta\delta_{r,f}) \end{cases} \tag{18}$$

where

$$\begin{cases} \bar{\rho}_r^s(i) = \rho_r^s(i) + b_{r,IF}(1) - b_{,IF}^s + k^s \Delta b_{r,IF}(1) \\ \tilde{I}_{r,1}^s(i) = I_{r,1}^s(i) + b_{r,GF}(1) - b_{,GF}^s + k^s \Delta b_{r,GF}(1) \\ \bar{\alpha}_{r,f}^s = \alpha_{r,f}^s + k^s \Delta\delta_{r,f} + \dfrac{\left[b_{,IF}^s - b_{r,IF}(1) - k^s \Delta b_{r,IF}(1)\right]}{\lambda_f^s} - \dfrac{\mu_f \cdot \left[b_{,GF}^s - b_{r,GF}(1) - k^s \Delta b_{r,GF}(1)\right]}{\lambda_f^s} \end{cases}$$

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